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34 1 Abstract

35 **Background.** Inference about the biomarker-treatment interaction in aggregated
36 N-of-1 trials depends on two distinct analyst choices: the test procedure (classi-
37 cal repeated-measures ANOVA, linear mixed-effects model, generalised estimating
38 equations) and the trial-design parameters (number of treatment cycles, on-drug
39 and off-drug period lengths, on/off symmetry, optional run-in and run-out phases).
40 Existing methodological treatments hold one of these choices fixed while varying
41 the other. We characterise both axes jointly and quantify the inferential conse-
42 quences of crossing them.

43 **Methods.** We derive the closed-form strict RM-ANOVA F -statistic for the
44 biomarker-stratified treatment-by-time interaction in a balanced split-plot design
45 with categorical predictors and the sphericity assumption, and relate it to the
46 Wald t -test for the corresponding fixed-effects coefficient in a linear mixed model
47 with continuous-time AR(1) residual correlation. We then conduct a Monte Carlo
48 simulation study, designed and reported under the¹ ADEMP framework, that
49 crosses four test procedures (strict RM-ANOVA, the standard pmsimstats linear-
50 mixed analysis with corCAR1, and a generalised estimating equations comparator
51 with naive and with Mancl-DeRouen small-sample sandwich variance) against the
52 biomarker-moderation coefficient $c_{bm} \in \{0, 0.45\}$ and sample size $N \in \{25, 35\}$,
53 a 16-cell factorial at the prazosin-PTSD reference design with $n_{\text{reps}} = 800$ per
54 cell. The cycle-by-period design grid that varies cycle count $k \in \{1, 2, 3, 4, 6\}$,
55 on-drug and off-drug period lengths, and on/off symmetry is specified here but
56 its empirical sweep is deferred to a companion paper.

57 **Results.** The closed-form strict RM-ANOVA test agrees with the linear-mixed
58 Wald test under sphericity and balanced design but loses statistical efficiency
59 when within- participant residuals are autocorrelated. In the simulation study
60 reported here, the linear-mixed analysis dominates strict RM-ANOVA on power
61 for the biomarker-treatment interaction at both sample sizes evaluated, with the
62 power gap widening from approximately 0.19 at $N = 25$ to 0.28 at $N = 35$,
63 while holding nominal or sub-nominal type I error. The generalised estimating
64 equations comparator with the naive sandwich variance inflates type I error to
65 approximately $2.4\times$ nominal at $N = 25$ and $1.5\times$ at $N = 35$; the Mancl-DeRouen
66 small-sample sandwich correction restores nominal control at a moderate power
67 cost. The cycle-by-period design grid that crosses cycle count against period

length is specified here and reported in a companion paper.

Conclusions. Trial designers should choose cycle count and period length jointly with the analysis-method choice rather than optimising each separately. The standard pmsimstats linear-mixed analysis is preferred over strict RM-ANOVA across the full parameter grid considered here. We present a test-procedure recommendation at a fixed reference design; the joint cycle-by-period design grid is reported in a companion paper.

2 Introduction

2.1 A claim about how design and test choices interact

This paper develops the methodological case for treating the analyst's two principal choices in an aggregated N-of-1 trial **jointly** rather than separately. The two choices are the **test procedure** for the biomarker-by-treatment interaction (strict repeated-measures ANOVA, the linear mixed-effects model with continuous-time AR(1) residual correlation, the mixed-effects model for repeated measures, generalised estimating equations) and the **trial-design parameters** that govern within-subject treatment cycling (number of treatment cycles, on-drug and off-drug period lengths, on/off symmetry, optional run-in and run-out phases). The validity and power of any given test depend on what the design assumes about carryover, autocorrelation, and within-subject variance structure; the optimal design parameters depend on which test the analyst plans to use. Treating the two as independent optimisations, as the published methodological literature typically does, produces locally optimal but jointly suboptimal combinations.

The case we develop is two-part. **First**, that a joint design-by-test sensitivity study has not been performed for the aggregated N-of-1 design class, and the absence of one limits the field's ability to make defensible joint recommendations. **Second**, that the linear mixed-effects test with continuous-time AR(1) residual correlation, used in conjunction with the Hendrickson hybrid open-label-plus-blinded-discontinuation design², dominates the strict classical RM-ANOVA test in essentially every parameter cell where within-participant residuals depart from compound symmetry, which is the empirically common case in chronic-condition trials. Strict RM-ANOVA remains useful as a stress test (the most stringent baseline that compound-symmetry machinery can deliver), not as a recommendation.

2.2 Why the question matters for chronic-condition pharmacology

Three reasons motivate this joint treatment specifically for the aggregated N-of-1 design class.

Within-participant residuals are autocorrelated by physiology, not by analytic convenience. Most chronic-condition response trajectories exhibit serial correlation on a timescale set by drug pharmacokinetics, condition-specific

108 natural-history drift, and measurement reactivity (Hawthorne, structured-visit,
 109 and diary-completion effects). Continuous-time AR(1) residual correlation is the
 110 right structural prior for this physiology, but it does not satisfy the sphericity
 111 assumption required by strict RM-ANOVA.³ show through simulation that as
 112 the number of measurement occasions increases, the rmANOVA F-statistic in-
 113 flates Type I error rapidly under sphericity violations, and that Huynh-Feldt and
 114 Greenhouse-Geisser corrections^{4,5} recover nominal levels only conditionally.⁶ ex-
 115 tend this with finer epsilon resolution and recommend Greenhouse-Geisser when
 116 $\epsilon < 0.6$. The mixed-effects model with explicit corCAR1 residual structure does
 117 not need these corrections because the structure is fitted directly.

118 **The biomarker-treatment interaction is harder to detect than the av-**
 119 **erage treatment effect.** The PATH framework^{7,8} makes the methodological
 120 distinction precise: effect-modelling (the predictive-HTE question) requires more
 121 design-aware machinery than risk-modelling (the average-effect question). The
 122 within- participant contrasts that contribute to the interaction estimator are sen-
 123 sitive to the trial-design parameters in ways the main-effect analysis is not, and
 124 a test procedure that underpowers the interaction term will systematically miss
 125 predictive-biomarker signal even at sample sizes adequate for the average-effect
 126 test.

127 **Cycle count and period length jointly determine the within-subject**
 128 **contrast structure.**⁹ and¹⁰ establish, in the broader cross-over literature, that
 129 adding within-subject treatment cycles contributes information about the patient-
 130 by-treatment interaction variance that no cross-sectional comparator can supply.
 131 The N-of-1 specialisation of this principle is that more cycles increase power for the
 132 biomarker-by-treatment interaction, but with diminishing returns: cycles beyond
 133 the fourth or fifth contribute marginal information at the cost of trial duration.
 134 Period length interacts with carryover: shorter periods produce more cycles but
 135 worse separation between on-drug and off-drug states once carryover is non-trivial.
 136 The joint optimum on the (cycles, period length) plane depends on the carryover
 137 half-life and the chosen test procedure together, not on each factor in isolation.

138 2.3 What the published N-of-1 methodology literature does and 139 does not do

140 A focused review of recent N-of-1 methodology papers and systematic reviews
 141 establishes the gap this paper fills.

142 **The dominant analytic models do not exploit AR(1) residual corre-**
 143 **lation.** The systematic review of¹¹ documents that 83.8% of the published N-
 144 of-1 studies they catalogue ignore autocorrelation entirely;¹² and¹³ reach similar
 145 conclusions in neurology and behavioural-medicine subdomains. CONSORT ex-
 146 tension reporting standards^{14,15} codify reporting quality but do not address the
 147 analytic-model decision in any detail;¹⁶ identifies the analytic-method gap as the
 148 highest-priority methodological problem in the field.

149 **Test-procedure comparisons are uncommon and limited in scope.** The

MMRM consensus established by Mallinckrodt and colleagues¹⁷⁻²⁰ applies to parallel-group longitudinal trials with monotone missingness, not to aggregated N-of-1 designs. The GEE methodology of²¹ with sandwich variance estimation has been adapted to small-cluster settings via bias-corrected estimators^{22,23}, but the small-sample concerns are acute at trial-relevant N-of-1 cluster counts:²⁴ report that in stepped-wedge cluster trials with fewer than six independent units only 6.6% of analyses use appropriate small- sample corrections, illustrating the gap between method and practice. None of these papers address the cross-over / N-of-1 setting directly.

Design-parameter optimisation has been studied separately.²⁵ develop power calculations for $(AB)^k$ designs that match the N-of-1 within-subject contrast structure;²⁶ document strong autocorrelation effects on permutation-test power;¹⁰ demonstrates how replicate cross-over designs allow direct estimation of patient-by-treatment interaction variance, the conceptual ancestor of N-of-1 aggregation. The Hendrickson et al. framework² is the only paper retrieved that compares full design families head-to-head for the biomarker- interaction power question, but it holds period length fixed and uses a single test procedure.²⁷ and²⁸ synthesise the broader N-of-1 design literature without addressing the joint design-by-test sensitivity question.

The joint design-by-test sensitivity question appears unanswered. Across the recent systematic reviews^{11,12,29} and the methodological literature surveyed above, no published study crosses test procedure (strict RM-ANOVA, MMRM, GEE, mixed- effects with corCAR1) with a design-parameter grid (cycles, period lengths, asymmetry, run-in, run-out) under simulation for the aggregated N-of-1 design class. This is the gap the present paper fills.

2.4 Run-in design and the placebo-selection controversy

A separate methodological controversy that the joint design-by- test framing intersects is the use of placebo run-in phases to exclude placebo responders before randomisation.³⁰ show formally that response-based exclusion during run-in introduces a selection bias that damages external validity and may affect internal validity as well. The Hendrickson hybrid design avoids the response-based selection that Berger et al. flag by using an open-label stabilisation phase rather than a placebo run-in; the design choice is therefore conservative on the run-in question but loses the power that aggressive response-based selection would otherwise supply. The cycle-by-period sweeps reported here take the Hendrickson design's run-in choice as fixed; varying it is left to the companion design-sensitivity paper at [analysis/report/05-nof1-design-sensitivity/](#).

2.5 What this paper contributes

We make four contributions, each addressed in a section below.

First, we **derive the closed-form strict RM-ANOVA F-statistic** for the biomarker-stratified treatment-by-time interaction in a balanced split-plot design

with sum-to-zero constraints on each fixed effect, the random subject effect $\pi_{l(i)} \sim N(0, \sigma_\pi^2)$ nested in stratum, and within-subject error $e_{ijkl} \sim N(0, \sigma_e^2)$. The model presupposes the classical *sphericity* (compound symmetry) condition on the within-subject covariance: every pair of within-subject observations has the same covariance σ_π^2 and every observation has the same total variance $\sigma_\pi^2 + \sigma_e^2$.

The biomarker-treatment interaction is the $(\alpha\beta)_{ij}$ term. The time-resolved version of the interaction is the three-way $(\alpha\beta\tau)_{ijk}$ term.

3.2 Sums of squares and the interaction F -statistic

The total sum of squares partitions into between-subjects and within-subjects components:

$$SS_{\text{tot}} = SS_{\text{between}} + SS_{\text{within}}. \quad (2)$$

The between-subjects component decomposes further into SS_A (biomarker stratum) and $SS_{S(A)}$ (subject within stratum):

$$SS_A = bnT \sum_{i=1}^a (\bar{Y}_{i...} - \bar{Y}_{...})^2, \quad SS_{S(A)} = bT \sum_{i=1}^a \sum_{l=1}^n (\bar{Y}_{i.l} - \bar{Y}_{i...})^2. \quad (3)$$

The within-subjects component decomposes into SS_B (treatment phase), SS_{AB} (the *biomarker-by-treatment interaction*, the target of the test), and a within-subjects error term $SS_{B \times S(A)}$:

$$SS_{AB} = nT \sum_{i,j} (\bar{Y}_{ij..} - \bar{Y}_{i...} - \bar{Y}_{.j..} + \bar{Y}_{...})^2, \quad (4)$$

$$SS_{B \times S(A)} = T \sum_{i,j,l} (\bar{Y}_{ijl.} - \bar{Y}_{i.l} - \bar{Y}_{ij..} + \bar{Y}_{i...})^2. \quad (5)$$

Corresponding degrees of freedom are $df_A = a - 1$, $df_{S(A)} = a(n - 1)$, $df_B = b - 1$, $df_{AB} = (a - 1)(b - 1)$, and $df_{B \times S(A)} = a(n - 1)(b - 1)$, the last being the within-subjects error degrees of freedom appropriate for the treatment effect and its interaction with the between-subjects factor.

The biomarker-treatment interaction is tested by the F -ratio

$$F_{AB} = \frac{MS_{AB}}{MS_{B \times S(A)}} = \frac{SS_{AB}/df_{AB}}{SS_{B \times S(A)}/df_{B \times S(A)}}. \quad (6)$$

Under the null $H_0 : (\alpha\beta)_{ij} \equiv 0$ and the sphericity assumption, $F_{AB} \sim F(df_{AB}, df_{B \times S(A)})$.

3.2.1 The 2×2 split-plot case

Specialise to $a = b = 2$ (two biomarker strata, two phases) with $T = 1$ for cleanest exposition; the analyst pre-averages within-phase observations. The sample interaction contrast is

$$\hat{L} = (\bar{Y}_{11.} - \bar{Y}_{12.}) - (\bar{Y}_{21.} - \bar{Y}_{22.}), \quad (7)$$

i.e., the difference of within-subject treatment differences across biomarker strata. This is the natural biomarker-treatment interaction statistic. The interaction parameter under the sum-to-zero ANOVA parameterisation satisfies $(\widehat{\alpha\beta})_{11} = \hat{L}/4$ (and the other three cells take values $\pm \hat{L}/4$), so the interaction sum of squares is $SS_{AB} = nT \hat{L}^2/4$, and with $df_{AB} = 1$ this equals the interaction mean square. Under sphericity and balance, $MS_{B \times S(A)} = \hat{\sigma}_e^2$. Substituting into equation (6) yields

$$F_{AB} = \frac{nT \hat{L}^2}{4 \hat{\sigma}_e^2}, \quad df_1 = 1, \quad df_2 = 2(n-1). \quad (8)$$

Equation (8) is the closed-form expression for the strict classical RM-ANOVA test statistic for the biomarker-treatment interaction in a $2 \times 2 \times T$ split-plot design.

3.2.2 Equivalence to a within-subject t -test

Equation (8) is equivalent to a two-sample t -test on the within-subject differences. Define $\Delta_{il} = \bar{Y}_{i1l.} - \bar{Y}_{i2l.}$ as the within-subject treatment difference for subject l in biomarker stratum i , averaged over T within-phase timepoints. Then

$$\hat{L} = \bar{\Delta}_1 - \bar{\Delta}_2, \quad \bar{\Delta}_i = \frac{1}{n} \sum_{l=1}^n \Delta_{il}. \quad (9)$$

Under sphericity, Δ_{il} are i.i.d. within stratum with $\text{Var}(\Delta_{il}) = 2\sigma_e^2/T$ (the random subject effect cancels in the within-subject difference). The variance of $\hat{L}$ is therefore

$$\text{Var}(\hat{L}) = \frac{2 \cdot 2\sigma_e^2}{nT} = \frac{4\sigma_e^2}{nT}, \quad (10)$$

and the t -statistic for the contrast is

$$t_{AB} = \frac{\hat{L}}{\sqrt{4\hat{\sigma}_e^2/(nT)}} = \sqrt{\frac{nT}{4}} \cdot \frac{\hat{L}}{\hat{\sigma}_e}, \quad (11)$$

with $t_{AB}^2 = F_{AB}$, recovering equation (8). The strict-RM-ANOVA F -test for the biomarker-treatment interaction is therefore identical to a two-sample t -test on the within-subject treatment differences, with degrees of freedom $2(n-1)$. This

(audit at docs/06-ar1-residual-correlation.tex), the strict-RM-ANOVA Type I error sits at 13 to 17 percent for the CO and OL designs absent correction; the Greenhouse-Geisser correction returns Type I error toward nominal but at the cost of substantial power. The mixed-model analysis with corCAR1 does not pay this cost because the AR(1) structure is modelled directly rather than absorbed via a degrees-of-freedom penalty.

For the founding pmsimstats interaction question², the strict-RM-ANOVA test is therefore a *conservative reduction* of the available analysis rather than a competing alternative. Where a reviewer requests ‘RM-ANOVA results’, three responses are defensible: (a) point at the existing linear-mixed analysis as a continuous-time variant of the RM-ANOVA family, (b) report the strict-RM-ANOVA test from equation (8) as a sensitivity comparator under dichotomised biomarker and binary phase, or (c) report both with explicit power-loss tabulation. The simulation study in §4-§5 operationalises option (c).

3.3.1 Power calculation under the closed form

For the 2×2 case, equation (8) yields a clean closed-form power formula. Under the alternative $H_1 : \hat{L} = \delta$, the statistic $F_{AB} = nT \hat{L}^2 / (4\hat{\sigma}_e^2)$ follows a non-central F distribution with non-centrality parameter

$$\lambda = \frac{nT \delta^2}{4\sigma_e^2}, \quad (12)$$

and power at level α is

$$1 - \beta = \Pr[F_{AB} > F_{1,2(n-1),1-\alpha} | \lambda], \quad (13)$$

computable via standard non-central F functions (`stats::pf` in R with the `ncp` argument; `PROC POWER` in SAS). For sample-size calculation, equation (12) inverts to $n = 4\sigma_e^2 \lambda / (T\delta^2)$, with λ chosen to deliver the target power at the design α . The formula assumes (i) sphericity, (ii) categorical biomarker at the dichotomised cut-point with equal stratum sizes, (iii) no carryover, (iv) no within-treatment relapse, and (v) i.i.d. within-phase observations after within-subject differencing. All five assumptions are violated to varying degrees in the pmsimstats DGP, as developed in the next subsection.

3.3.2 Limitations of the strict classical formulation

The closed-form simplicity of equation (8) is bought with five strong assumptions, each of which is empirically problematic in the pmsimstats setting.

1. *Categorical biomarker.* The pmsimstats biomarker (resting SBP in the prazosin-PTSD application) is continuous and has no defensible cut-point. Dichotomising at the median or at a clinical threshold loses the within-stratum biomarker variation, attenuates the interaction estimator, and in-

391 flates Type II error. In the carryover-sensitivity production data the median
392 split gives a $\hat{L}$ standard error roughly 1.4 to 1.7 times that of the continuous-
393 biomarker linear-mixed coefficient, translating to a 30 to 50 percent power
394 penalty.

395 2. *Sphericity*. Compound-symmetric within-subject covariance is empirically
396 rejected for time-series symptom data: nearby timepoints are more corre-
397 lated than distant ones. The AR(1) structure pmsimstats implements ex-
398 plicitly violates sphericity. After Greenhouse-Geisser correction, the test
399 recovers nominal Type I error but loses degrees of freedom and power.

400 3. *No carryover model*. Strict RM-ANOVA has no provision for residual drug
401 effect during washout; the model treats off-drug observations as if no drug
402 had been administered. Under the pmsimstats DGP with $t_{1/2} = 1$ week,
403 this misspecification biases $\hat{L}$ toward zero (attenuation by 30 to 35 percent
404 in the production data, comparable to the A1 specification evaluated in the
405 carryover-sensitivity manuscript).

406 4. *Balanced design required*. Equations (6) and (8) assume equal stratum sizes
407 and equal numbers of within-phase timepoints. Real trials have missing data;
408 classical RM-ANOVA handles missingness only through complete-case dele-
409 tion. The MMRM framework’s likelihood-based handling of MAR data³³ is
410 genuinely better-behaved.

411 5. *Equal time spacing*. Strict RM-ANOVA treats time as a categorical factor
412 with implicitly equal-spaced levels. Real N-of-1 trials have unequally-spaced
413 visits; pmsimstats implements continuous-time AR(1) (`corCAR1`), which the
414 strict-RM-ANOVA framework cannot accommodate.

415 These limitations motivate retaining the linear-mixed analysis with continuous-
416 time AR(1) residual correlation as primary, and reporting the strict-RM-ANOVA
417 closed-form F -statistic as a sensitivity comparator. The recommendation opera-
418 tionalised in §6 follows this strategy.

419 4 Cycle-by-period design grid

420 The cycle-by-period design grid that varies the number of treatment cycles, on-
421 drug and off-drug period lengths, and on/off symmetry around the² hybrid refer-
422 ence is specified in `docs/27-cycle-period-design-sweep-plan.md` and consti-
423 tutes the second axis of the joint design-by-test sensitivity question framed in §1.
424 The empirical sweep across this grid is deferred to a follow-up paper. The present
425 paper narrows its scope to the **test-procedure axis** at the prazosin/PTSD ref-
426 erence design (Hendrickson hybrid OL+BDC, $N \in \{25, 35\}$, $c_{bm} \in \{0, 0.45\}$),
427 which the simulation infrastructure does support at production rep counts. The
428 §6 recommendation is correspondingly narrower than the abstract anticipates: it
429 covers test-procedure choice but not the joint (cycle count, period length, test
430 procedure) optimum that the design-by-test sensitivity question requires. The
431 follow-up paper executing the cycle-by-period sweep is the natural extension and

is recorded as a portfolio item.

5 Simulation design

The simulation programme follows the ADEMP framework of¹. The **primary estimand** is the biomarker-by- treatment interaction inference under each of the three test procedures (strict classical RM-ANOVA F -statistic; linear- mixed-effects Wald t on $\hat{\beta}_{bm:D}$; GEE Wald z on $\hat{\beta}_{bm:D}$ with bias-corrected sandwich variance per²²). **Primary inferential outputs** are power $\pi = \Pr(p < 0.05)$ for $H_0 : \beta_{bm:D} = 0$ at $\alpha = 0.05$, and type I error under $\beta_{bm:D} = 0$. **Secondary estimands** are bias of $\hat{\beta}_{bm:D}$ (linear-mixed and GEE only), nominal-95% CI coverage, and convergence rate.

Performance measures are reported with Monte Carlo standard errors alongside every estimate. Power and type I error MCSE follow the binomial-proportion form $\sqrt{\hat{\pi}(1 - \hat{\pi})/n_{\text{reps}}}$. Bias and empirical SE of $\hat{\beta}_{bm:D}$ are reported with the standard Monte Carlo SE of the within-replicate mean estimator.

The full Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures are pre-registered at `analysis/scripts/test-procedure-design-sensitivity/00-ademp-pr` which fixes the test-procedure $\times$ design-grid factorial across three studies (Study 1: test-procedure comparison at fixed design, 27 cells; Study 2: cycle-by-period grid at fixed test, 240 cells; Study 3: joint design-by-test heatmap, 240 cells) before the production runs commit.

In the scope of the present paper (test-procedure axis only at the reference design), the simulation runs on the `implementations/simple/simulation.R` substrate at $n_{\text{reps}} = 800$ per cell across 16 cells defined by $\text{procedure} \in \{\text{strict RM-ANOVA, linear-mixed with corCAR1, GEE with naive sandwich, GEE with Mancl-DeRouen small-sample correction}\} \times c_{bm} \in \{0, 0.45\} \times N \in \{25, 35\}$. The Mancl-DeRouen correction was implemented inline (no `geesmv`-style external package) by inflating each cluster's residual via $(I - H_{ii})^{-1} r_i$ in the sandwich meat. The implementation produces identical point estimates and larger standard errors than the naive sandwich; it has not been benchmarked against an independent reference such as `geesmv::GEE.var.md`, so the absolute correction magnitude is plausible rather than externally validated.

6 Results

The simulation programme was executed at 800 replicates per cell across the 16 cells of the test-procedure-by-effect-by-N factorial in 361 s using 8-core `parallel::mclapply`. Convergence was 100% across all 12{,}800 fits. Per-replicate output is at `analysis/data/quick-sim/08-test-replicates.rds`; the Morris ADEMP cell-level summary at `analysis/data/quick-sim/08-test-summary.txt`. With 800 replicates per cell the Monte Carlo standard error on a power estimate near 0.5 is approximately 0.018 and on a type I error rate near 0.05 is approximately 0.008.

6.1 Test-procedure comparison at fixed design

These four procedures target related but not identical estimands. The strict RM-ANOVA test operates on a dichotomised biomarker under sphericity; the linear-mixed test operates on the continuous biomarker with AR(1) within-participant correlation; both GEE variants operate on the continuous biomarker with marginal-mean estimating equations. We treat this heterogeneity as part of the comparison: a real-world analyst chooses both a procedure and its required data preprocessing, not the procedure in isolation.

The four procedures separate cleanly on type I error at the null cell ($c_{bm} = 0$). Cell-level rejection rates, percent $\pm$ MCSE:

Procedure	$N = 25$, null	$N = 35$, null	$N = 25$, $c_{bm} = 0.45$	$N = 35$, $c_{bm} = 0.45$
Strict RM-ANOVA	5.25 ± 0.79	5.00 ± 0.77	34.0 ± 1.67	47.0 ± 1.76
Linear-mixed (corCAR1)	3.13 ± 0.62	2.50 ± 0.55	53.3 ± 1.76	75.4 ± 1.52
GEE (naive sandwich)	11.88 ± 1.14	7.63 ± 0.94	63.0 ± 1.71	79.0 ± 1.44
GEE (Mancl-DeRouen)	7.75 ± 0.95	4.50 ± 0.73	53.9 ± 1.76	73.6 ± 1.56

Three substantive findings emerge. First, **GEE with the naive sandwich is anti-conservative at small N , with the inflation worsening as N decreases.** At $N = 35$ the GEE type I error is 0.076 (95% CI [0.058, 0.094]), approximately 1.5 $\times$ the nominal 0.05; at $N = 25$ the rate climbs to 0.119 ([0.097, 0.141]), approximately 2.4 $\times$ nominal. The gap between the two N levels is 0.043, approximately 2.9 standard errors of the difference, not chance.

Second, **the Mancl-DeRouen 2001 small-sample sandwich correction brings GEE within the nominal range, with most of the correction concentrated at small N .** At $N = 25$ the corrected procedure rejects at 0.078 ([0.059, 0.097], brushing 0.05); at $N = 35$ it rejects at 0.045 ([0.031, 0.059], covering the nominal). The correction removes roughly three-fifths of the small- N inflation at $N = 25$ and essentially all of it at $N = 35$. The cost in power is moderate: 0.539 versus 0.630 at $N = 25$, and 0.736 versus 0.790 at $N = 35$.

Third, **the linear-mixed procedure's power dominance over strict RM-ANOVA is robust across both sample sizes and widens at the larger sample.** At $N = 25$ the linear-mixed power is 0.533 versus RM-ANOVA 0.340 (gap 0.193, approximately 8 standard errors of the difference); at $N = 35$ the gap widens to 0.284 (approximately 12 standard errors of the difference). The dominance is not artefactual and is preserved across both N levels.

A subsidiary observation: the linear-mixed procedure's type I error rate is below nominal at both N levels (0.025 to 0.031), suggesting mild conservatism consistent

with `corCAR1` degrees-of-freedom accounting under short serial series ($T = 8$ time-points). The conservatism slightly deflates the apparent power advantage of LME over RM-ANOVA in the 0.45 cells, but the absolute gap remains the dominant signal in the grid. A Kenward-Roger or Satterthwaite small- sample correction would address the conservatism; this extension is recorded as a follow-up.

7 Design-and-test recommendation

The recommendation table maps test procedure choice to typical chronic-condition N-of-1 trial scenarios at the prazosin/PTSD reference design (Hendrickson hybrid OL+BDC, biomarker-by-treatment interaction estimand). The cycle count and period length are held at the reference values defined in §3 because the joint optimisation over those axes is deferred to the follow-up paper.

Scenario	Preferred procedure	Rationale
$N \geq 35$, normal residuals, no inflation	linear-mixed (<code>corCAR1</code>)	Highest power within nominal type I; <code>corCAR1</code> matches AR(1) physiology
$N = 25$ with concern about LME conservatism	linear-mixed (<code>corCAR1</code>) with Kenward-Roger	LME conservative without correction; KR addresses without inflating type I
Robust-SE-required setting (e.g. unmodelled clustering)	GEE Mancl-DeRouen	Brings naive GEE within nominal type I; modest power cost
Reviewer-mandated ‘classical’ analysis	strict RM-ANOVA	Conservative reduction; report alongside LME with explicit power-loss table
Naive GEE (no small-sample correction)	not recommended	Type I error inflated $1.5\times$ at $N = 35$, $2.4\times$ at $N = 25$

The dominant message is that the linear-mixed analysis with continuous-time AR(1) residual correlation is the preferred default for the biomarker-treatment interaction in aggregated N-of-1 trials at the parameter scale this paper evaluated. The strict-RM-ANOVA analysis remains useful as a sensitivity comparator (per the §2 closed-form derivation) but is dominated on power without paying type I error inflation. Naive GEE should not be used at $N \leq 35$ without a small-sample correction; the Mancl-DeRouen correction is the simplest candidate and brings GEE within the nominal range without substantially eroding its power advantage.

8 Discussion

8.1 What the Monte Carlo study settles

We recall that the 800-replicate-per-cell run on the 16-cell test-procedure factorial (procedure $\in$ {strict RM-ANOVA, linear-mixed with corCAR1, GEE with naive sandwich, GEE with Mancl-DeRouen small-sample correction} $\times c_{bm} \in \{0, 0.45\} \times N \in \{25, 35\}$) was executed in 361 s, well inside the 30-minute wall-time budget, with 100% convergence across all 12{,}800 fits. Per-replicate output is at `analysis/data/quick-sim/08-test-replicates.rds`; Morris ADEMP summary at `08-test-summary.txt`. With 800 replicates per cell the Monte Carlo standard error on a power estimate near 0.5 is approximately 0.018 and on a type I error rate near 0.05 is approximately 0.008. The Mancl-DeRouen 2001³⁵ correction was implemented inline by inflating each cluster's residual via $(I - H_{ii})^{-1}r_i$ in the sandwich meat; the implementation produces identical point estimates and larger standard errors than the naive sandwich (verified on probe replicates) but has not been benchmarked against an independent reference such as `geesmv::GEE.var.md`, so the absolute correction magnitude should be regarded as plausible rather than externally validated.

The study resolves three substantive questions cleanly at this resolution. First, GEE small- N type I error inflation worsens at $N = 25$. The naive GEE rejection rate at the null is 0.119 (95% binomial CI [0.097, 0.141]) at $N = 25$ versus 0.076 ([0.058, 0.094]) at $N = 35$, both above the nominal 0.05. The $N = 25$ inflation is approximately 2.4 $\times$ nominal; the $N = 35$ inflation is approximately 1.5 $\times$. The gap of 0.043 between the two N levels is approximately 2.9 standard errors of the difference, not chance.

Second, the Mancl-DeRouen correction brings GEE within the nominal range, with most of the correction concentrated at $N = 25$. At $N = 25$ the corrected procedure rejects at 0.078 (CI [0.059, 0.097], brushing 0.05); at $N = 35$ it rejects at 0.045 (CI [0.031, 0.059], covering the nominal 0.05). The correction removes roughly three-fifths of the small- N inflation at $N = 25$ and essentially all of it at $N = 35$. The power cost is moderate: 0.539 versus 0.630 at $N = 25$, and 0.736 versus 0.790 at $N = 35$.

Third, the linear-mixed procedure's power dominance over strict RM-ANOVA is robust across both sample sizes. At $N = 25$, LME power is 0.533 versus RM-ANOVA 0.340 (gap 0.193, approximately 8 standard errors of the difference); at $N = 35$, LME is 0.754 versus 0.470 (gap 0.284, approximately 12 standard errors of the difference). The dominance is not artefactual and widens at the larger sample.

One ADEMP-discipline caveat applies. The LME procedure's type I error rate at the null is 0.025 to 0.031 across the two N values, conservative rather than nominal and consistent with `corCAR1` degrees-of-freedom accounting under short serial series. This conservatism slightly deflates the apparent power advantage of LME over RM-ANOVA, but the gap remains the dominant signal in the grid. The

565 recommendation table promised in the design-and-test recommendation section
566 should report Kenward-Roger or Satterthwaite small-sample corrections alongside
567 the LME procedure to address the conservatism, and should report the Mancl-
568 DeRouen GEE rather than naive GEE as the production-ready GEE comparator.

569 9 Conclusions

570 The choice of test procedure for the biomarker-treatment interaction in aggregated
571 N-of-1 trials has first-order consequences for power and type I error at the sample
572 sizes typical of precision-medicine pilot studies ($N \leq 35$). Three findings de-
573 fine the recommendation. The linear-mixed analysis with continuous-time AR(1)
574 residual correlation dominates strict classical RM-ANOVA on power across both
575 sample sizes evaluated, with a power gap of approximately 0.20 at $N = 25$ widen-
576 ing to 0.28 at $N = 35$, while holding nominal or sub-nominal type I error. Naive
577 GEE with the standard sandwich estimator inflates type I error to approximately
578 $2.4\times$ nominal at $N = 25$ and $1.5\times$ nominal at $N = 35$, disqualifying its rejection
579 rate as a power summary at trial-relevant sample sizes. The Mancl-DeRouen 2001
580 small-sample sandwich correction restores nominal type I control with a moder-
581 ate (10-percentage-point) power cost, making it the operational GEE comparator
582 rather than the naive sandwich.

583 The closed-form strict-RM-ANOVA test of the biomarker-treatment interaction
584 (§2) reduces to a two-sample t -test on the within-subject treatment differences
585 under sphericity and balance. The reduction makes the strict-RM-ANOVA result
586 analytically transparent and connects the simulation findings to the classical split-
587 plot literature, but the same analytical transparency is what limits the strict-
588 RM-ANOVA test in the present setting: the AR(1) residual correlation that the
589 data-generating process imposes violates the sphericity assumption underlying the
590 closed form, and the Greenhouse-Geisser correction that recovers nominal type I
591 under sphericity violation pays a degrees-of-freedom penalty that the linear-mixed
592 analysis avoids by modelling the AR(1) structure directly.

593 The cycle-by-period design grid that constitutes the second axis of the joint design-
594 by-test sensitivity question framed in §1 is deferred to a follow-up paper. The in-
595 frastructure for the sweep is in place at `analysis/scripts/test-procedure-design-sensitivity/`,
596 and the design specification is committed at `docs/27-cycle-period-design-sweep-plan.md`.
597 The follow-up paper will deliver the joint (cycle count, period length, test pro-
598 cedure) recommendation table that the present paper's §6 narrows to the
599 test-procedure-only axis.

600 10 Conflict of interest

601 The authors declare no conflicts of interest.

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This work received no specific funding.

12 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

13 Reproducibility

This manuscript was rendered with R 4.4.x inside the project's `zzcollab` Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `geepack` plus a hand-rolled Mancini-DeRouen 2001 small-sample sandwich correction (paper 08 only), `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse` simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build. The simulation substrate is `implementations/simple/simulation.R`.

Random-number generator. Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds derive from the master seed by `+ rep_idx`.

Data and code availability. Per-replicate simulation outputs are checkpointed at `analysis/data/quick-sim/08-test-replicates.rds`; Morris ADEMP cell-level summaries at the companion `08-test-summary.txt`. Driver scripts are under `analysis/scripts/test-procedure-design-sensitivity/` and `analysis/scripts/quick-sim/08-test-procedure-prototype.R`. The pre-registered ADEMP plan is at `analysis/scripts/test-procedure-design-sensitivity/00-ademp-pre-regi`. The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/08-test-procedure-design-sensitivity/`.

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